



< Previous							Next >
-------------------------------	---	---	---	---	---	---	---------------------------

1.2.1 Absolute value

 Bookmark this page

Video

Advanced Linear Algebra: Foundations to Frontiers

1.2.1 Absolute value

Robert van de Geijn
Maggie Myers

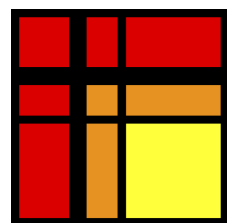


 [Download video file](#)

 [Download SubRip \(.srt\) file](#)

 [Download Text \(.txt\) file](#)

1/1 point (graded)



Read Unit 1.2.1 of the notes. [\[LINK\]](#)

☒ done



Submit

✓ Correct (1/1 point)

Hide Discussion

Topic: Week 01 / 01.2.1

Add a Post

Show all posts 

by recent activity ▼

☒ Why are Complex Numbers Considered Scalars?

5

Since a complex number has two orthogonal components, why is it considered a scalar rather than a vector?

Homework 1.2.1.1

3/3 points (graded)
Evaluate the below.

Example: Enter $3 - 2i$ as 3 - 2 * i .

1.

$(1 + i)(2 - i)$
=

3+i

✓ Answer: 3+i
2.

$(2 - i)(1 + i)$
=

3+i

✓ Answer: 3+i
3.

$\overline{(1 - i)}(2 - i)$
=

3+i

✓ Answer: 3+i
4.

$\overline{(1 - i)} =$
 $(2 - i)$

3-i

✓ Answer: 3-i
5.

$\overline{(2 - i)}(1 - i)$
=

3-i

✓ Answer: 3-i
6.

$(1 - i)\overline{(2 - i)}$
=

3-i

✓ Answer: 3-i

Solution:

For a complete solution see the [notes](#).

Submit

Answers are displayed within the problem

Homework 1.2.1.2

1/1 point (graded)
Let $\alpha, \beta \in \mathbb{C}$.

$\alpha\beta = \beta\alpha$

ALWAYS

✓ Answer: ALWAYS

$\overline{\alpha}\beta = \overline{\beta\alpha}$

SOMETIMES

✓ Answer: SOMETIMES

Solution:

For a complete solution see the [notes](#).

Submit

Answers are displayed within the problem

Homework 1.2.1.3

1/1 point (graded)
Let $\alpha, \beta \in \mathbb{C}$.

$$\overline{\alpha\beta} = \overline{\beta\alpha}$$

ALWAYS

✔ Answer: ALWAYS

Solution:

For a complete solution see the [notes](#).

Submit

📘 Answers are displayed within the problem

Homework 1.2.1.4

1/1 point (graded)
Let $\alpha \in \mathbb{C}$.

$$\overline{\alpha}\alpha \in \mathbb{R}.$$

ALWAYS

✔ Answer: ALWAYS

Solution:

For a complete solution see the [notes](#).

Submit

📘 Answers are displayed within the problem

Homework 1.2.1.5

1/1 point (graded)
The absolute value function is homogeneous. That is, $|\alpha\beta| = |\alpha||\beta|$ for all $\alpha, \beta \in \mathbb{C}$.

TRUE

✔ Answer: TRUE

Now prove it!

Solution:

For a complete solution see the [notes](#).

Submit

📘 Answers are displayed within the problem

Homework 1.2.1.6

1/1 point (graded)
Let $\alpha \in \mathbb{C}$.

$$|\overline{\alpha}| = |\alpha|.$$

ALWAYS

✔ Answer: ALWAYS

Now prove it!

Solution:

For a complete solution see the [notes](#).

Submit

i Answers are displayed within the problem

< Previous

Next >

© All Rights Reserved



edX

- [About](#)
- [Affiliates](#)
- [edX for Business](#)
- [Open edX](#)
- [Careers](#)
- [News](#)

Legal

- [Terms of Service & Honor Code](#)
- [Privacy Policy](#)
- [Accessibility Policy](#)
- [Trademark Policy](#)
- [Sitemap](#)
- [Cookie Policy](#)
- [Your Privacy Choices](#)

Connect

- [Idea Hub](#)
- [Contact Us](#)
- [Help Center](#)
- [Security](#)
- [Media Kit](#)



© 2023 edX LLC. All rights reserved.
深圳市恒宇博科技有限公司 [粤ICP备17044299号-2](#)